

A market study of the Thai fisheries sector, illustrated by shrimp aquaculture.

— Valorization of marine resources —



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1) Country overview: Thailand

Thailand is a country located in Southeast Asia, on the Indochinese Peninsula, which also includes Vietnam, Laos, and Cambodia. Thailand shares borders with five countries: Laos and Cambodia to the east, Myanmar to the west, Malaysia to the south, and the Andaman Sea to the west.

Thailand stretches over more than 800 kilometers from east to west and 1,770 kilometers from north to south, with a total area of 513,120 km². Extending nearly 2,000 kilometers in length, the country experiences a variety of climates. Overall, Thailand has two main seasons: a dry season from December to April and a monsoon season from May to November, with average annual temperatures ranging between 25°C and 35°C.

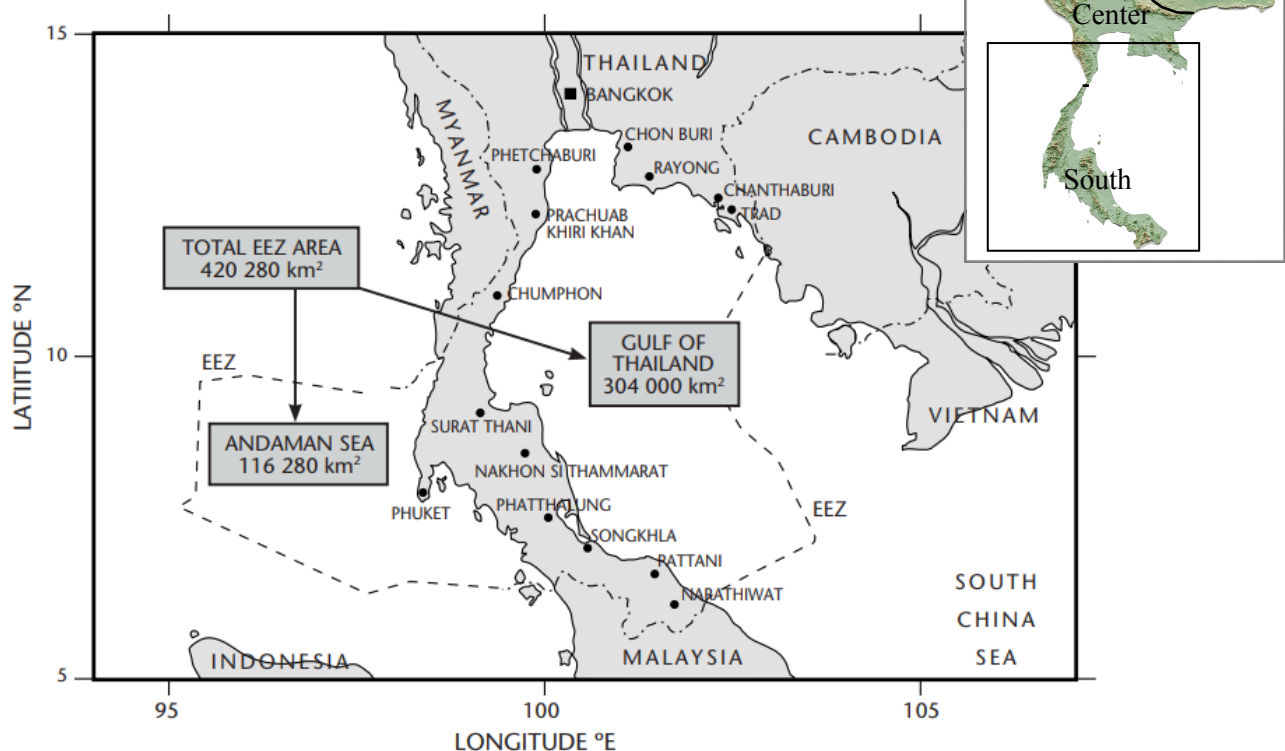


Figure 1: Topographic map of Thailand (Janekitkosol et al., 2003)

Thailand has a population of 71 million inhabitants, of whom around 11 million live in the capital, Bangkok. Its gross domestic product (GDP) reached 495 billion US dollars in 2022 (World Bank), with an annual growth rate of approximately 2.6% that same year. GDP per capita was 6,909 US dollars in 2022, compared to 40,886 US dollars in France in the same year. Thailand is the second-largest economy in Southeast Asia, after Indonesia, it remains a developing country. Its economy is highly dependent on tourism and exports. In 2018, Thailand exported a total of 249.8 billion US dollars worth of goods, with 12% of these exports going to China and 11% to the United States.

Thailand's coastlines are highly popular with tourists, particularly the Andaman Sea (located in the southwest of the country, in the Indian Ocean) and the Gulf of Thailand in the southeast. This strong tourist appeal is explained by its idyllic landscapes, turquoise waters, and fine sandy beaches. In addition, the exceptional biodiversity found in these waters, along with the wide range of

activities available (kayaking, diving, snorkeling, etc.), significantly contributes to the tourism economy.

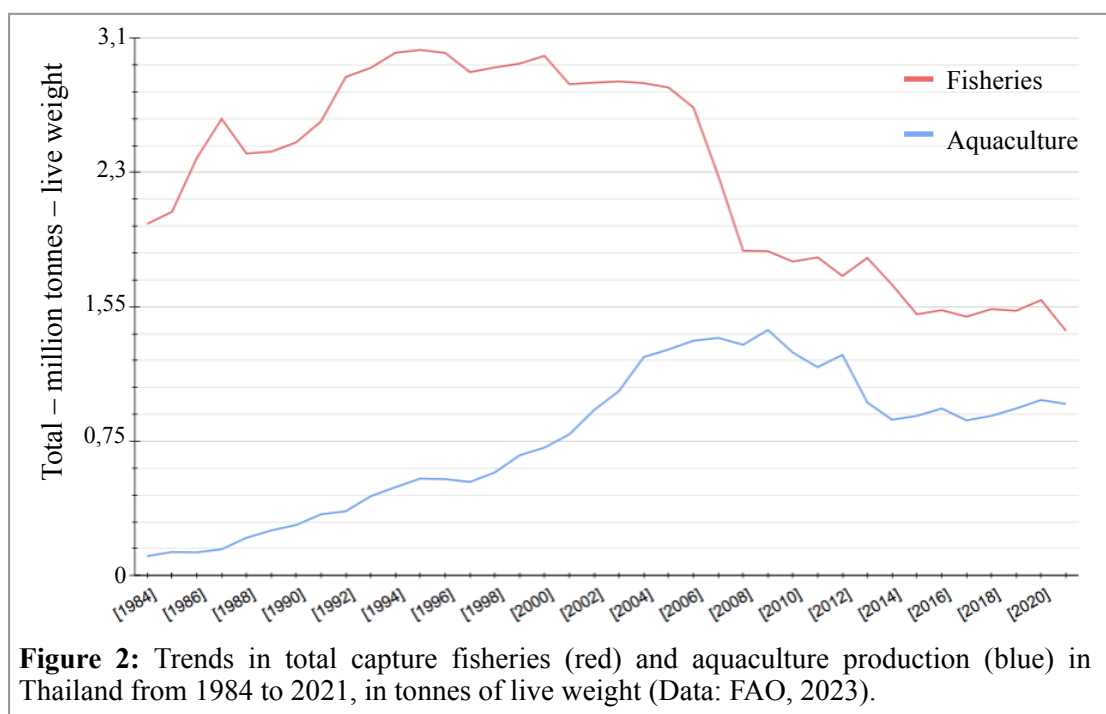
Approximately 13% of Thailand's territory is protected through 12 national parks and 44 nature reserves, 18 of which are marine areas encompassing islands and mangroves. More than 600 species inhabit these waters, including reef fish, sea turtles, seabirds, and many others. Tourism accounted for around 15% of GDP in 2018, representing a value of 74.2 billion US dollars (French Treasury).

Beyond tourism, these coastlines have also enabled the country to develop its industrial sector through the construction of ports facilitating both imports and exports. Fishing has thus become a major activity, both economically and culturally. Thailand's Exclusive Economic Zone (EEZ) covers 420,280 km², including 304,000 km² in the Gulf of Thailand and 116,280 km² in the Andaman Sea (Figure 1; Janekitkosol et al., 2003; Stobutzki et al., 2006).

2) Analysis of production and its recent trends

Over the past decade, the Thai government has implemented regulations aimed at promoting more sustainable fishing practices. Prior to 2015, Thai fisheries operated largely under an open-access regime, which led to overfishing and the collapse of fish stocks, thereby threatening the country's economy (FAO, 2001). The stagnation in total catches observed between 1992 and 2004 reflects the weakening of these stocks, which had been subject to overexploitation since the 1960s (Figure 2). Despite increased fishing effort, stock levels were insufficient to sustain higher catches. The sharp decline in catches observed in 2005 is therefore the result of both stock collapse and the devastating impact of the 2004 tsunami on the Thai fishing fleet (**Figure 2**).

In 2006, Thailand was accused of exploiting child labor within its fishing industry. As a result, major importers of Thai seafood products, including the United States and the European Union, cancelled contracts with numerous Thai retailers and processors, contributing to a decline in



demand and, consequently, a decrease in production (Prompatanapak & Lopetcharat, 2020; **Figure 2**).

In the 1970s, Thailand began developing its aquaculture sector, initially focusing on one fish species, three species of bivalves, and shrimp of the genus *Penaeus* (**Figures 3A** and **4**). Over time, aquaculture expanded to include a wider variety of species produced in larger quantities (Sampantamit et al., 2020). Mollusks long accounted for around half of Thailand's aquaculture production, until the country shifted its focus toward the shrimp market (**Figure 3B**). Today, mollusk farming represents approximately 10% of total production, compared to 45.7% for fish and 44% for crustaceans (Sampantamit et al., 2020).

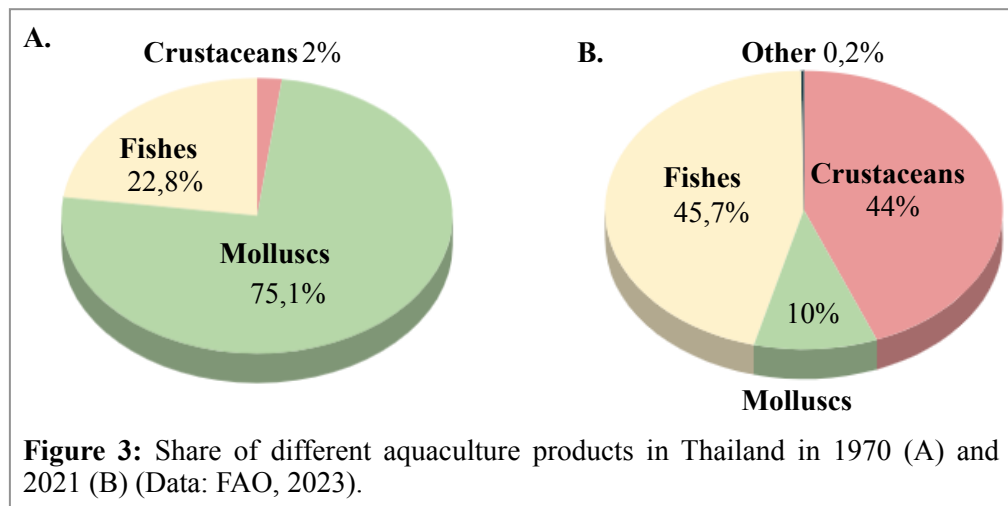


Figure 3: Share of different aquaculture products in Thailand in 1970 (A) and 2021 (B) (Data: FAO, 2023).

Thai aquaculture was severely affected by the emergence of a new disease in 2012: Early Mortality Syndrome (EMS), which caused high mortality rates in shrimp and led to a decline in aquaculture production (**Figure 2**). As a result, the cost of Thai seafood products became significantly higher than that of competing countries. Due to all the events and factors mentioned above, Thai seafood production has declined over the past 15 years (Prompatanapak & Lopetcharat, 2020).

In 2020, however, Thailand remained the twelfth-largest producer in terms of capture fisheries, with 1.58 million tonnes of live weight caught (FAO, 2020). In the same year, the country produced around 1 million tonnes through aquaculture, making it the sixth-largest producer of crustaceans in marine and coastal aquaculture.

3. Analysis of the distribution of production by fisheries and/or aquaculture products

Until the 1960s, fishing in Thailand was predominantly artisanal, focusing on the capture of small pelagic species such as mackerel and anchovies. These fisheries relied on small vessels and mainly supplied local markets. In the early 1960s, a German–Thai project was launched, introducing the use of light trawlers in the Gulf of Thailand to target demersal species. The expansion of trawling led to a significant decline in catches due to overfishing, with yields decreasing from around 50 kg per hour in the 1980s to only 26 kg per hour in the 1990s (Pauly et al., 2003).

Over time, Thailand developed a multi-gear fishery, enabling it to exploit a wide range of species. Total catches consist of approximately 33% pelagic fish, 30% bycatch, 18% demersal fish, 7% cephalopods, and 4% crustaceans (Panjarat, 2008; FAO, 2023).

When freshwater fish are included, accounting for around 8% of total live weight production, marine fish represent nearly 75% of Thailand's total live weight, followed by freshwater fish (8%), molluscs (7.8%), and crustaceans (7.1%) (**Figure 4**; FAO, 2023).

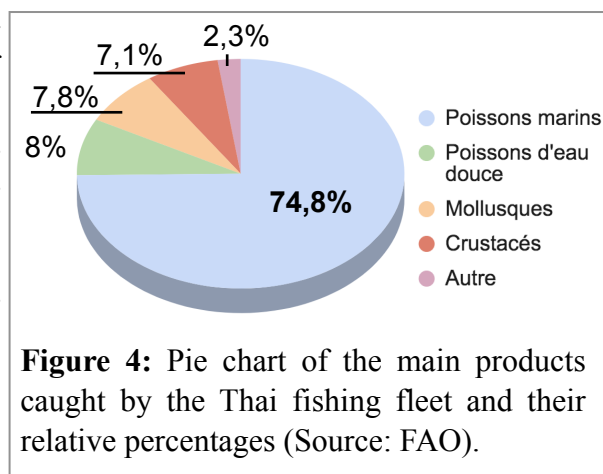


Figure 4: Pie chart of the main products caught by the Thai fishing fleet and their relative percentages (Source: FAO).

Marine fish catches are predominantly composed of pelagic species, including anchovies (25%), sardines (18%), and mackerel (17%) (**Figure 5**). Since the 1980s, sardine catches have declined sharply (−32%), partly offset by an increase in anchovy catches (+20%). Sardine landings have shown significant fluctuations over recent decades, partly due to inadequate stock management. In contrast, anchovy catches have remained relatively stable, supported by the development of fishing gear and techniques, such as light lures, which have greatly enhanced their capture (FAO, 2023; Panjarat, 2008).

As a result, pelagic resources have been fully exploited, and in some cases overexploited, since the 1980s. This is particularly true for anchovies, mackerel, small tunas, and flatfish. Consequently, in 2020,

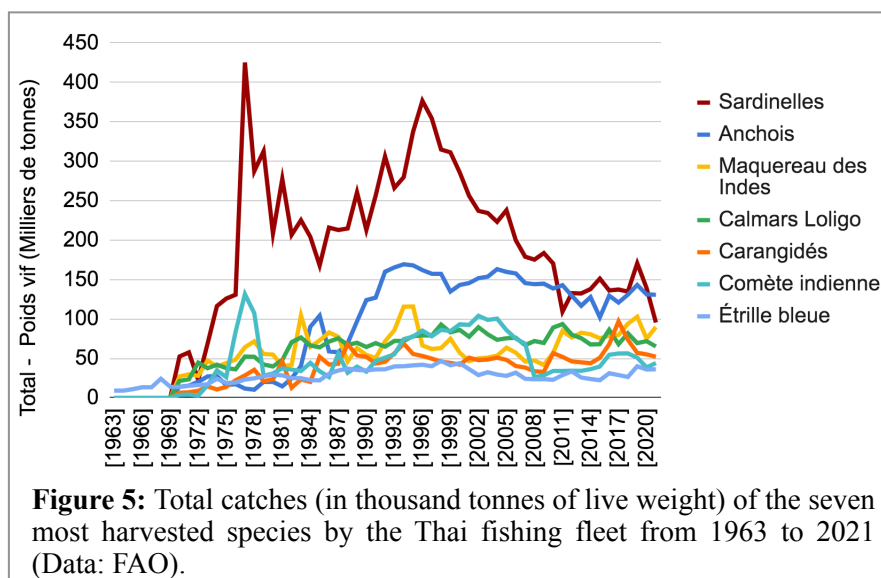


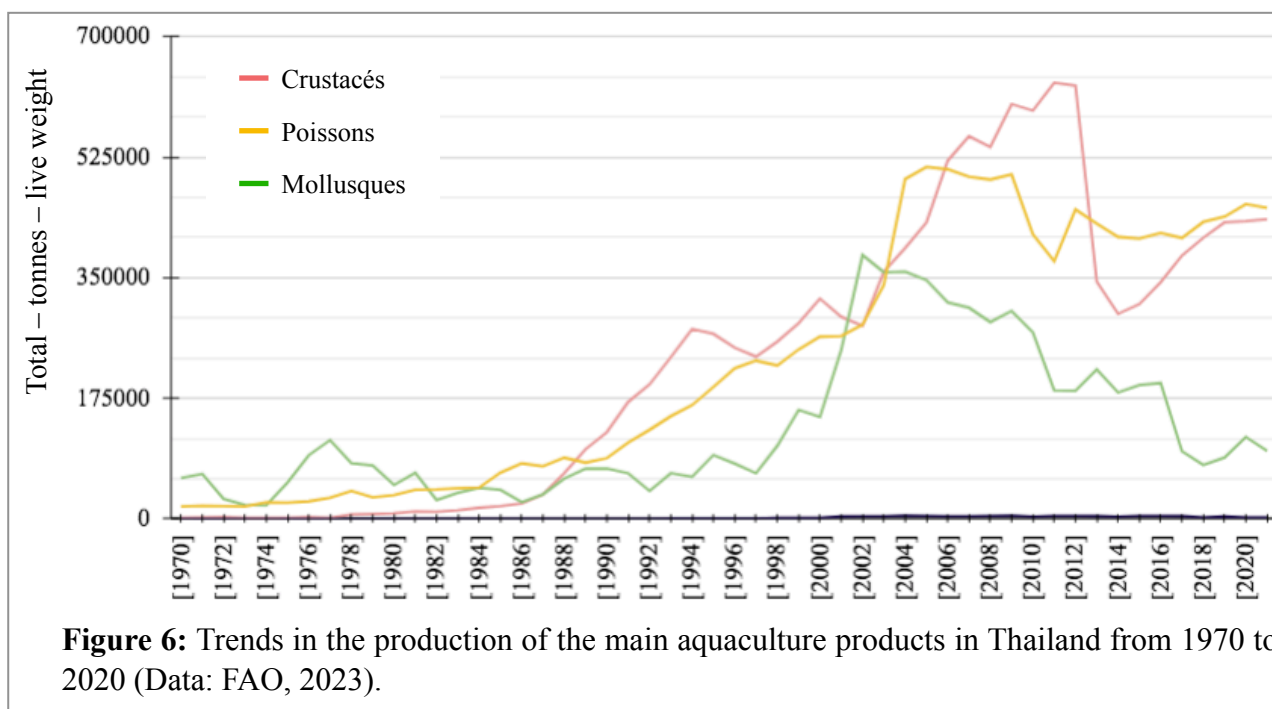
Figure 5: Total catches (in thousand tonnes of live weight) of the seven most harvested species by the Thai fishing fleet from 1963 to 2021 (Data: FAO).

Thailand ranked as the sixth-largest fishing nation for the group comprising herrings, sardines, and anchovies, and the tenth-largest producer of cephalopods (squid and octopus) (Ministry of Agriculture, 2023).

As for crustaceans (shrimp and crabs), they account for only 5% of marine catches but represented 21% of total value in 2004. In Thai waters, penaeid shrimp (*Penaeus* spp.) and small-sized shrimp have been fully overexploited, which partly explains the development of shrimp aquaculture.

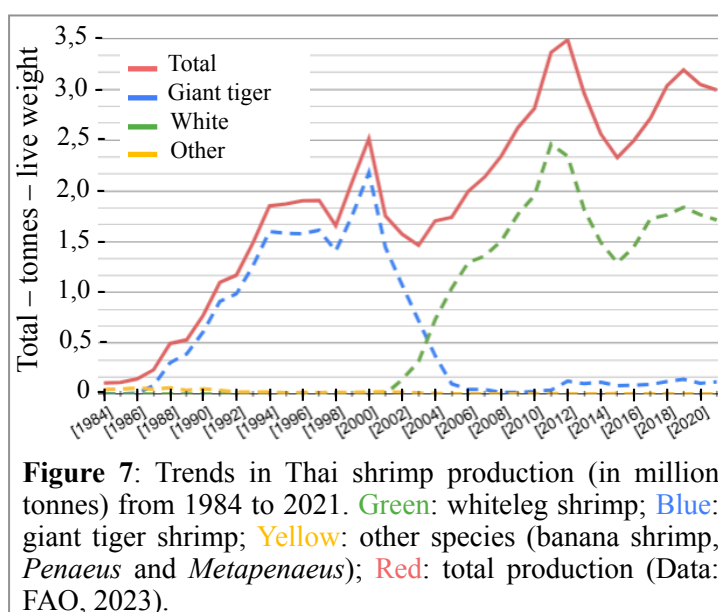
Thai aquaculture experienced significant growth from the late 1980s onward, particularly through the expansion of crustacean and fish farming (**Figure 6**). It can be divided into two main categories: inland freshwater aquaculture and coastal/marine aquaculture. In 2020, approximately 62% of total production (617,900 tonnes) came from coastal aquaculture, while the remaining 38% (384,600 tonnes) was produced by freshwater aquaculture.

The three main aquaculture products in Thailand are crustaceans, fish, and molluscs. Farmed crustaceans have been the leading source of aquaculture production, accounting for about 40% of total output, of which 95% originates from coastal aquaculture. Fish represent 38% of aquaculture production (96% from freshwater systems), while molluscs account for 22% (FAO, 2022).



Mollusc aquaculture also experienced significant growth in the early 2000s, although this expansion was short-lived and remained about half as productive as crustacean farming. Since the decline in aquaculture in 2012, fish farming has become the dominant sector, reaching 452,000 tonnes in 2021, compared to 435,000 tonnes for shrimp and 99,000 tonnes for molluscs (FAO, 2023).

With regard to shrimp aquaculture, Thailand was one of the first countries to develop this sector in the 1970s and was a global leader in shrimp farming prior to the 2000s. The most widely cultivated species was the giant tiger shrimp (*Penaeus monodon*), also known as Black Tiger shrimp. However, the sector was severely impacted in the 1990s by the emergence of two viral diseases affecting shrimp: Yellow Head Virus (YHV) and White Spot Syndrome Virus (WSSV), leading to stagnation and even a decline in production (Prompatanapak & Lopetcharat, 2020; **Figure 7**).



To counter this decline in productivity, Thailand replaced Black Tiger shrimp with whiteleg shrimp (*Litopenaeus vannamei*), a species resistant to YHV and WSSV. This shift contributed to an increase in both the volume and value of Thai exports between 2001 and 2004 (Prompatanapak & Lopetcharat, 2020; Sampantamit et al., 2020).

The second major decline in production observed in 2012 was caused by the emergence of Early Mortality Syndrome (EMS) (see Section 2), which can result in mortality rates of up to 100% in affected populations, leading to a 50% drop in shrimp production (Prompatanapak & Lopetcharat, 2020; FAO, 2020; **Figures 6 and 7**).

Thailand, once a global leader in shrimp aquaculture, was severely affected by a series of disease outbreaks that decimated its shrimp populations. As a result, the country has never fully recovered the levels of production and prosperity it previously achieved in this sector.

4. Links between fisheries and the population's food situation, with a One Health analysis

Between 1990 and 2021, the average per capita consumption of seafood doubled. In 2019, it was estimated that seafood provided 20% of protein intake for half of the world's population and 40% for the Thai population, which consumes on average 30.5 kg of fish per person per year (Tasse, 2024; World Bank, 2020; FAO, 2006; Panjarat, 2008; Stobutzki et al., 2006).

Thailand has more than 2,500 fishing villages along the Gulf of Thailand and the Andaman Sea. Among these fishers, 80% practice artisanal fishing passed down through generations. Fish represent a crucial source of both food and income for the Thai population (FAO, 2006). Thailand is bordered by the sea to the south, east, and west, and also possesses an extensive freshwater network, particularly rivers. As a result, fishing is a highly accessible activity for coastal populations and those living near water bodies.

Government policies implemented in recent years have led to a significant reduction in the poverty rate (6.2% in 2020) as well as in the prevalence of undernourishment (5% in 2020) (World Bank, 2020). One of Thailand's key strengths is its food self-sufficiency, which enables it not only to feed its population but also to export large quantities of food, such as rice and seafood products from fisheries and aquaculture.

However, despite these efforts, significant disparities remain in access to resources between urban, coastal, and rural populations. Infrastructure and distribution networks are far more developed in urban and coastal areas, while rural populations are more vulnerable to poverty and food insecurity. In addition, rural households, often dependent on agriculture, are highly exposed to external factors such as weather conditions.

Seafood products are particularly valuable due to their nutritional composition, being rich in omega-3 fatty acids, vitamins, and minerals. They provide an essential source of protein and other nutrients. Furthermore, the aquaculture and fisheries sectors generate thousands of jobs and contribute significantly to the country's GDP.

However, ensuring product quality is essential to maintain a healthy population capable of sustaining economic activity. This requires the implementation of strict quality controls, particularly in aquaculture production. Contaminated organisms containing heavy metals, chemicals, or pathogens could pose serious risks to public health. Moreover, healthy farmed organisms are less susceptible to diseases (viral and bacterial), which are a major concern in aquaculture.

Similarly, inadequate control of fishing quotas can lead to stock depletion and negatively impact the fisheries sector, as observed in the early 2000s with the collapse of fish stocks due to overfishing in the absence of effective regulation. In this context, aquaculture may represent a viable solution to meet the nutritional needs of the Thai population while preserving marine ecosystems.

Animal health has become a major concern, particularly following the emergence of shrimp viruses that severely impacted shrimp aquaculture in the 1990s. Organisms in poor condition tend to face low consumer demand, as they are often less appealing and have lower nutritional value. In addition, some diseases can be transmitted from animals to humans (zoonoses); therefore, any diseased organism is considered unfit for consumption, leading to significant economic losses for

aquaculture producers. Developing a more sustainable and responsible aquaculture sector would thus represent a major advantage for the country, ensuring higher-quality food production while preserving the environment.

Environmental health remains fragile in Thailand, as this concept has only recently gained attention. Following the collapse of fish stocks in the early 2000s, the government gradually introduced regulations in fisheries and aquaculture. The implementation of fishing licenses, quotas, and size regulations has allowed fish stocks to progressively recover. However, aquaculture has expanded since the 1970s with limited consideration of its environmental impacts. As a result, mangroves have been degraded, and freshwater and coastal ecosystems have been polluted. Furthermore, in the absence of effective regulation and sanitary control, shrimp farms have been severely affected—and in some cases devastated—by diseases such as Yellow Head Virus (YHV) and White Spot Syndrome Virus (WSSV) (see Section 3). Poor environmental health leads to reduced seafood production and contributes to higher levels of poverty and undernourishment, limiting access to a key source of protein in the country.

In conclusion, fisheries and aquaculture play a crucial role in Thailand, both for the national economy and for food security. Animal, environmental, and human health are closely interconnected, which justifies the use of a One Health approach. Although progress has been made, significant improvements are still needed, particularly in terms of environmental health. While regulations have been introduced, their enforcement remains insufficient in some cases. Strengthening controls on fishing quotas, product quality, and disease monitoring systems would improve environmental protection and ensure higher-quality food for consumers.

Another important issue concerns labor conditions in the fisheries sector. Thailand has previously been criticized for poor working conditions in the seafood industry. Forced labor and child labor have been reported, accounting for up to 26% of workers on vessels involved in fishing for animal feed products. Although recent data remain limited, the government has implemented stricter policies to address these issues (Nakamura et al., 2018).

5. Market organization (prices, purchasing, sales)

At the national level:

Thailand's seafood supply chain can be divided into six main segments: (1) primary producers (fisheries and aquaculture), (2) intermediaries (fish markets), (3) processors, (4) distributors (exporters and importers), (5) retailers (supermarkets), and (6) consumers (Prompatanapak & Lopetcharat, 2020). Following the numerous transitions and disruptions that occurred between 1990 and 2004 (see Section 2), profit margins declined, while processing costs, quality control requirements, and certification expenses increased. As a result, some traders and fish exporters were excluded from the supply chain. After concerns regarding labor conditions in the Thai fishing industry were raised in 2006, all workers within the supply chain are now required to be legally registered with the Ministry of Labour of the Kingdom of Thailand under the Labour Protection Act B.E. 2560 (Ministry of Labour, 2017; Prompatanapak & Lopetcharat, 2020). In addition, labor costs in Thailand (accounting for around 20% of total costs) are significantly higher than in neighboring countries, with a minimum wage of approximately US\$9.85 per day compared to US\$5.75 in Vietnam (Prompatanapak & Lopetcharat, 2020). Consequently, Thai seafood products have become less competitive than those of other Asian producers.

At the global level:

At the global scale, Thailand ranks among the most competitive seafood-exporting countries (**Figure 8**). Although export volumes remain significant, they have been surpassed by import volumes since 2015. This increase in imports is partly explained by the development of the tuna canning industry in Thailand. Despite a decline in export volumes, exports still generate substantially higher revenues than imports. In 2019, Thailand imported approximately 2 million tonnes of seafood products and exported 1.9 million tonnes, representing values of 3.77 billion US dollars and 5.86 billion US dollars, respectively. This indicates that although import volumes exceed export volumes, the value of exported products is significantly higher than that of imported products.

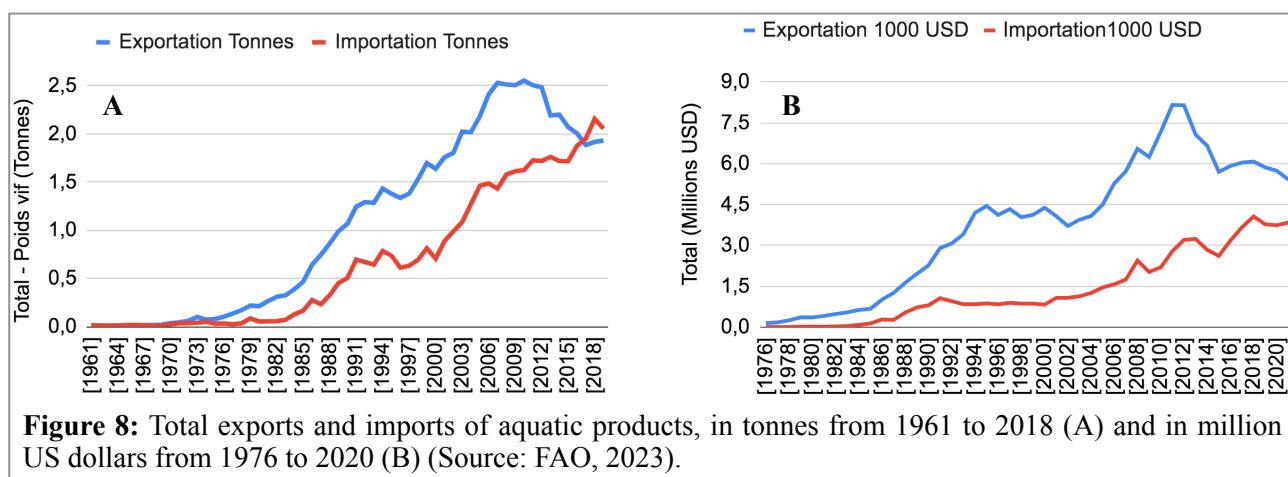


Figure 8: Total exports and imports of aquatic products, in tonnes from 1961 to 2018 (A) and in million US dollars from 1976 to 2020 (B) (Source: FAO, 2023).

Regarding imported products, 70% consist of pelagic fish (including tuna), 10% cephalopods, 8.6% freshwater fish, 4.7% molluscs, and 3.9% crustaceans (FAO, 2023). The volume of imports has increased in recent years, largely driven by the development of the tuna canning industry, which requires the importation of approximately 80% of the tuna processed in Thailand (Ministry of Agriculture, 2023).

Thailand's main exports consist of prepared fish and crustacean products, with tuna alone accounting for a value of 2.2 billion euros. In addition, exports of unprocessed fish, crustaceans, and molluscs reached 1.6 billion euros in 2020, including 0.7 billion euros from shrimp alone. These figures contributed to Thailand accounting for 3.5% of global seafood exports (Ministry of Agriculture, 2023; FAO, 2023; Kulanujaree et al., 2020).

The main exported products are pelagic fish (62%), including tuna. Tuna is imported, processed, and then re-exported to consumer countries. The tuna canning industry is dominated by Thai Union Group, which holds 37% of the market, followed by Sea Value with 15%. Thailand has been considered a global leader in this sector in recent years.

The remaining exports consist of crustaceans (16.2%), a high proportion explained by shrimp aquaculture, followed by molluscs (7%), demersal fish (6%), freshwater fish (5.4%), and cephalopods (2.8%) (FAO, 2023).

Thailand is also one of the leading exporters of aquaculture products in Asia. In 2018, the country exported a total of 249.8 billion US dollars worth of goods, with 12% of exports going to China and 11% to the United States. Thailand is therefore one of the world's largest shrimp exporters, with its main markets being the United States, Europe, and Asia (particularly China).

In international trade, shrimp is the most widely farmed group of species. Global production reached nearly 7.4 million tonnes in 2021, with a value of approximately 44.7 billion US dollars

(Miao & Wang, 2020; FAO, 2023). In 2018, shrimp accounted for 64% of global farmed crustacean production, with a total value of 39 billion US dollars (FAO, 2020). In 2021, global shrimp exports and imports were valued at 29.7 billion and 28.5 billion US dollars, respectively (FAO, 2023).

As shrimp aquaculture is a highly profitable sector driven by strong global demand, other countries have also expanded their shrimp farming industries, leading to intense competition and a decline in demand for Thai products. Although Thailand was the global leader in shrimp production until the early 2000s, it has since been overtaken by countries such as China, Indonesia, India, and Vietnam. In 2021, Thailand and China produced approximately 388,000 tonnes and 1.2 million tonnes of shrimp, respectively, with values of 1.8 and 17.4 billion US dollars.

China is currently the world's largest producer of farmed shrimp (Table 1) and has significantly increased its share of global production, rising from 13.2% in 1998 to 34.2% in 2018 (Miao & Wang, 2020). Over the past

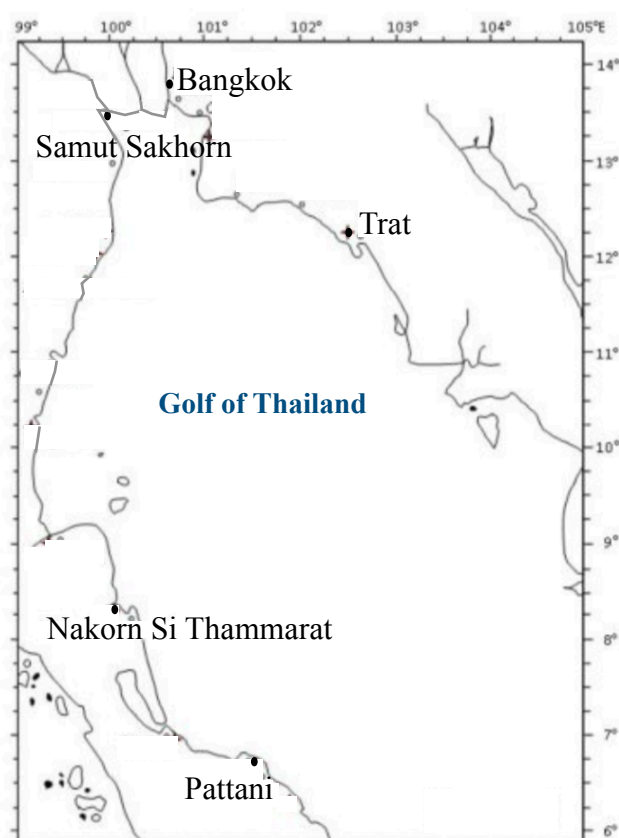


Figure 9: Map of the main landing ports in the Gulf of Thailand (FAO, 2006).

	Production (1000 tonnes)		Valeur de la production (millions \$ USD)	
	1991	2021	1991	2021
Chine	220	2 319	1 537	17 452
Inde	40	1 036	168	5 362
Indonésie	136	941	878	4 407
Thaïlande	162	388	948	1 832
Viet Nam	36	985	165	7 171

Table 1: Production volume (thousand tonnes) and production value (million US dollars) of shrimp among leading shrimp-farming countries (Source: FAO, 2023).

decade, the market share of Thailand and Ecuador—formerly among the leading global producers—has declined by more than 50%. Meanwhile, Vietnam, India, and Indonesia have considerably increased their share of global shrimp production between 1998 and 2018.

Although Thailand remains one of the major shrimp-producing countries, a significant gap has emerged between the global leader (China) and Thailand. In 2021, China produced nearly 1.9 million tonnes more shrimp than Thailand, representing a difference of approximately 15.6 billion US dollars between the two countries (Table 1; FAO, 2023). Furthermore, countries such as India

and Vietnam, which produced four times less shrimp than Thailand in 1991, now produce nearly three times more (Table 1).

6. Characterization of the country's fishing model, including the number of fishers and vessels

Thailand operates a multi-gear, multi-species fishery, similar to other Asian countries. Fishing vessels vary in size and capacity, ranging from large commercial trawlers to small-scale artisanal boats. In 2015, a total of 42,512 vessels were recorded, of which 78% were artisanal (33,208 vessels), while the remaining 22% were classified as commercial vessels (Stobutzki et al., 2006). Fishing activities within Thailand's Exclusive Economic Zone (EEZ), located in the Gulf of Thailand and the Andaman Sea, account for approximately 60% of total marine catches (FAO, 2008).

The Gulf of Thailand accounts for 27,507 fishing vessels (18,933 artisanal and 8,574 commercial), while the Andaman Sea hosts 8,657 vessels, including 7,342 artisanal and 1,315 commercial boats (Kulanujaree et al., 2020). Despite the higher number of artisanal vessels, commercial vessels dominate production, accounting for around 90% of total marine catches in Thailand (FAO, 2008).

Commercial vessels are capable of undertaking multi-day fishing trips at sea. They primarily use medium to large trawls (41%), purse seines (11%), surrounding gillnets (14%), and drift nets (FAO, 2008; Stobutzki et al., 2006). In contrast, artisanal vessels rely more on small trawls, gillnets, traps, as well as lines and longlines. Thailand historically operated an intensive fishing system until fish stocks collapsed. In response, measures have been implemented to transition toward a more sustainable and environmentally friendly model of fisheries and aquaculture. In total, more than twenty different fishing gears are used in the Thai fisheries sector (Kulanujaree et al., 2020; FAO, 2011; FAO, 2008).

In 2006, the main Thai landing ports included Songkhla, Pattani, Samut Sakhon, Nakhon Si Thammarat, and Trat (**Figure 9**; FAO, 2006).

In 2020, the fisheries sector provided employment to 172,430 fishers, of whom 82% were migrants (Kulanujaree et al., 2020). Similarly, in the aquaculture sector, it is estimated that in 2010, around 80% of shrimp production originated from small-scale family farms (FAO, 2011).

7. Impact of national fisheries and aquaculture policies on their development

In Thailand, fisheries management measures have been implemented to rebuild fish stocks depleted by overfishing. These measures fall under the "Thai Fisheries Act" and include regulations on fishing zones, gear types, and the number of vessels authorized to operate (FAO, 2005).

Since 1984, certain fishing areas have been seasonally closed, prohibiting the use of trawls and seines with mesh sizes smaller than 4.7 cm. This regulation aims to protect juvenile fish and short mackerel, an economically important species. Additional regulations apply within the 3,000-meter coastal zone, particularly concerning dredging activities, which are highly destructive to seabeds and demersal species. In the 1980s, measures were also introduced to limit the number of fishing licenses and better control fishing capacity. However, these measures proved ineffective due to insufficient enforcement by the relevant authorities (FAO, 2011).

Fisheries management in Thailand was reformed in 2015 through the Royal Ordinance on Fisheries. Prior to 2015, Thai fisheries operated under an open-access system, which led to overexploitation and stock collapse, threatening the country's economy (FAO, 2001). There were no effective limits on the number of vessels or fishing effort (Kulanujaree et al., 2020; Charles, 2002). This lack of regulation resulted in a decline in catch per unit effort (CPUE) in both the Gulf of Thailand and the Andaman Sea between 1961 and 2019.

In 2015, the Thai Department of Fisheries (DOF) committed to aligning with the United Nations Sustainable Development Goals. To this end, Thailand introduced catch limits aimed at maintaining fish stocks at sustainable levels and ending overfishing (FAO, 2016). Analyses based on both holistic and analytical models identified key issues, including the capture of juveniles, stock depletion, the use of inappropriate fishing gear, and non-compliance with protected areas and breeding seasons. However, despite the implementation of these measures, no significant improvement in fish stocks was observed, largely due to inadequate and poorly enforced licensing systems (Kulanujaree et al., 2020).

At the same time, Thailand implemented measures to combat illegal, unreported, and unregulated (IUU) fishing, with the aim of developing a more sustainable and socially responsible seafood industry (Kulanujaree et al., 2020; FAO, 2001). A licensing system was introduced, and authorities were assigned to conduct inspections. However, the multi-gear and multi-species nature of Thai fisheries made enforcement particularly challenging. As a result, Thailand adopted regulations based on fishing effort, as well as limits on the number and size of vessels.

Regarding aquaculture, Thailand has recently adopted a more sustainable and less expansion-driven approach. The Fisheries Act prohibits the establishment of new aquaculture farms in mangrove areas, although existing farms are allowed to continue operating. Shrimp farmers are permitted to construct ponds only on their own land and within coastal regions, in order to prevent saltwater intrusion into soils and groundwater. In addition, farms larger than 8 hectares are now required to register and obtain operating permits. Environmental impact assessments may also be required for such farms (FAO, 2011; FAO, 2006).

More broadly, environmental policies across aquaculture are based on the FAO Code of Conduct. Farms must obtain environmental compliance certification, and monitoring of chemical and pharmaceutical use is required and provided free of charge to producers. These measures aim to meet HACCP (Hazard Analysis Critical Control Point) standards imposed by importing countries (FAO, 2011). Overall, these policies have proven to be effective in recent years.

8. Regional impact of fisheries and/or aquaculture on coastlines and ports (with a map), including a One Health analysis

Mangroves are unique ecosystems located at the interface between tidal zones and coastal regions, characterized by vegetation primarily composed of mangrove trees. These trees and shrubs are adapted to brackish water conditions, where freshwater mixes with seawater in shallow environments. Mangroves host a high level of biodiversity, including species such as plankton, algae, molluscs, crustaceans, and fish, as well as various visitors such as snakes, lizards, turtles, and numerous bird species that feed in the mudflats (Sampantamit et al., 2020; WWF France, n.d.). They provide a wide range of ecosystem services, including habitat for many species, food

resources for humans, and protection against coastal erosion and flooding, among others (Sampantamit et al., 2020). Mangroves also serve as nursery grounds for many species, particularly wild shrimp. It has been shown that one hectare of mangrove forest can support between 100 and 1,000 kg per year of marine fish and shrimp catches (De Graaf & Xuan, 1998; Hambrey, 1996). Shrimp farming in mangrove areas therefore leads to the transformation of an ecosystem providing multiple public services into a privately used, single-purpose system (Bailey, 1988).

According to official reports from the Department of Fisheries (DoF), between 1995 and 2015, there were on average 40,884 coastal aquaculture farms annually, covering approximately 77,800 hectares. Of these farms, 67% were dedicated to shrimp farming, 20% to fish farming, and 13% to shellfish farming (Sampantamit et al., 2020). It is estimated that Thailand lost approximately 122,300 hectares of mangroves over half a century (1961–2014) (**Figure 10**).

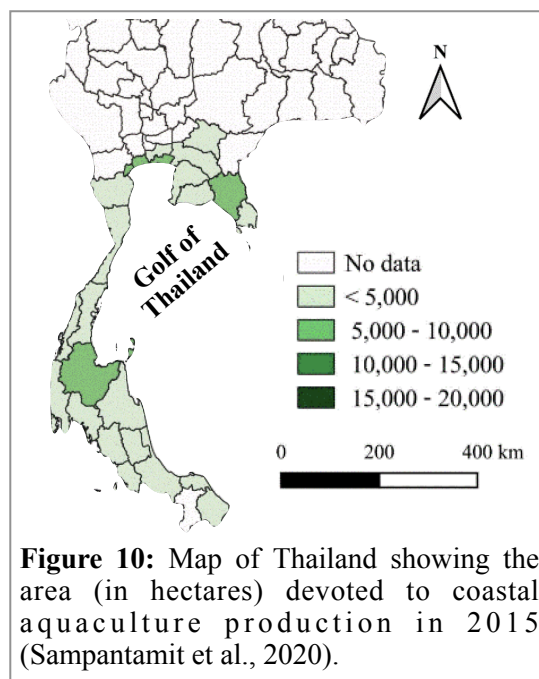


Figure 10: Map of Thailand showing the area (in hectares) devoted to coastal aquaculture production in 2015 (Sampantamit et al., 2020).

The economic consequences of coastal ecosystem degradation include coastal erosion, saltwater intrusion into groundwater and agricultural lands, and a reduction in the availability of valuable goods and ecosystem services derived from mangroves and other coastal wetlands (De Graaf & Xuan, 1998). Moreover, the conversion of mangroves into aquaculture areas results in greenhouse gas emissions ranging from 1,067 to 3,003 megagrams of CO₂ equivalent (CO₂e) per hectare. Consequently, the carbon footprint of shrimp production—estimated at 1,603 kg CO₂e per kilogram—exceeds that of beef (1,440 kg CO₂e per kilogram) (Boone Kauffman et al., 2017). At the same time, the accumulation of aquaculture waste, such as nitrogen, can lead to environmental acidification and cause harmful and potentially irreversible effects on aquatic plants and animals, as well as on coral ecosystems, resulting in significant biotic damage (Sampantamit et al., 2020).

The One Health concept places health at the center of the interactions between the environment, humans, and animals. From a human perspective, aquaculture offers significant nutritional benefits and is expected to supply the majority of marine-derived protein by 2050. Fish, crustaceans, and molluscs are essential sources of protein, nutrients, and minerals for human health. In Thailand, seafood products meet a substantial share of the population's protein requirements. However, meeting such demand requires maximizing production efficiency, which in turn highlights the importance of animal health.

To ensure maximum yield and profitability, aquaculture ponds are often overcrowded, and animals are poorly fed with low-quality feed, resulting in a reduction in their nutritional value. Overstocking in these systems promotes the rapid spread of diseases (bacterial, viral, and parasitic), leading to the excessive use of antibiotics and other treatments. Thailand provides a relevant example of the close interconnection between animal and human health. As previously mentioned, Thai aquaculture has been severely affected by multiple disease outbreaks causing high mortality in shrimp populations. Overstocked ponds have facilitated the rapid transmission of diseases, leading to the collapse of entire production units. This high mortality has resulted in reduced shrimp

production, leading to a decline in seafood availability, lower incomes for farmers, and broader negative impacts on the national economy. In addition, the use of pharmaceutical treatments to control these diseases may pose risks to human health, including the development of antibiotic-resistant bacteria and the presence of antibiotic residues in aquaculture products intended for human consumption.

From an environmental perspective, intensive aquaculture contributes significantly to the degradation of coastal ecosystems, particularly mangroves, due to water pollution, the use of chemicals in ponds, and the destruction of natural habitats. Furthermore, the conversion of natural ecosystems into aquaculture areas can lead to biodiversity loss, including the destruction of nursery grounds, disruption of trophic networks, and the introduction of non-native species (through escapes from aquaculture facilities).

Finally, the conversion of mangroves into aquaculture farms reduces the capacity for atmospheric carbon sequestration, thereby indirectly contributing to climate change, ocean acidification, and sea-level rise.

In summary, aquaculture plays a key role in supplying populations with seafood, which is an important source of proteins and essential minerals. However, the quality of these products can easily be compromised by the excessive use of growth hormones and antibiotics, with direct consequences for public health. Moreover, the destruction of mangroves, which contributes indirectly to climate change, leads in particular to sea-level rise, threatening millions of coastal households. In addition, forced labor remains a significant issue in aquaculture, particularly in Asian countries, highlighting the need for governments to implement measures ensuring fair and safe working conditions. Adopting a One Health approach in Thai aquaculture could promote more sustainable practices that balance human, animal, and environmental health, thereby strengthening the resilience of coastal ecosystems. Nevertheless, significant challenges remain in Thailand across human, animal, and environmental dimensions.

9. SWOT analysis (Strengths, Weaknesses, Opportunities, Threats)

Thailand's tropical climate enables the development of highly diversified aquaculture, allowing the country to offer a wide range of aquatic products, including shrimp, tilapia, and catfish, thereby effectively meeting the demands of international markets. Its food self-sufficiency also allows Thailand to innovate and invest in improving its export capacity. The development of the tuna canning industry is a clear example. Recognizing its limited competitiveness in tuna fisheries, Thailand chose instead to specialize in processing. The country imports raw tuna, processes it, and re-exports it to meet growing global demand. This processing sector provides access to high-quality resources at significantly lower costs than fresh products. In addition, canned products are easy to store, which represents a major advantage, particularly for developing countries.

However, increasing competition, inflation, and Thailand's dependence on external suppliers for raw materials have reduced the profitability of both fishers and aquaculture producers. At the same time, environmental degradation has led to declining catches, further exacerbating economic vulnerability within these sectors. To compensate for these losses, producers tend to increase output—fishing more and intensifying aquaculture practices—which in turn places additional pressure on the environment. This creates a self-reinforcing negative cycle.

Developing more sustainable and higher-quality fisheries and aquaculture practices could help break this cycle. Moreover, Thailand appears to have reached a plateau in production capacity and is unable to compete with major producers such as China, Japan, India, and Indonesia.

Therefore, shifting toward a quality-oriented market rather than a quantity-driven model would be a strategic approach. As global demand for high-quality products continues to rise, promoting responsible aquaculture and sustainable fisheries could allow Thailand to differentiate itself and better align with its production capabilities.

		Strengths	Weaknesses
Internal	Natural environment	. Country surrounded by the sea on multiple sides . Extensive network of freshwater bodies and coastal waters suitable for aquaculture . High coastal biodiversity . Tropical climate ideal for aquaculture development . Strategic geographic location (proximity to other Asian countries)	. Destruction of coastal habitats due to overfishing and aquaculture (e.g., mangrove destruction)
	Production	. Production of a wide range of aquaculture species . Modern aquaculture techniques improving production efficiency and product quality . Multi-gear fishing practices	. High vulnerability of production to diseases (e.g., shrimp viruses) . Production largely focused on shrimp farming . Dependence on imported raw materials for fisheries and aquaculture . Declining or insufficient wild broodstock for aquaculture
	Market	. Geographical proximity reducing transportation costs . Growing domestic demand for protein	. Increasing costs of labor and raw materials
	Regional disparities	. Well-developed coastal infrastructure supporting fisheries and aquaculture	. Restricted market access for rural areas
	Occupations	. Traditional fisheries provide a skilled and experienced workforce, with knowledge passed down through generations	. Difficult working conditions, low wages, and high risk of accidents . Commercial vessels, though limited in number, account for a large share of total catches, creating strong competition for small-scale fishers . Competition between farmers and aquaculture producers for access to water resources
	Policies	. Government policies supporting and promoting sustainable and environmentally friendly practices	. Weak enforcement of fisheries regulations . Insufficient quality control of aquaculture products

		Opportunities	Threats
External	Natural environment	. Potential for aquaculture development (rivers and mangroves) . Adoption of sustainable practices to protect the environment and restore fish stocks	. Climate change impacts organism physiology (e.g., delayed reproduction in aquaculture)
	Production	. Development of methods and technologies to maximize production while improving resource management . Use of recirculating aquaculture systems (RAS) to preserve freshwater resources and marine ecosystems . Modern processing, packaging, and port infrastructure enabling improved storage and export capacity	. Environmental: extreme weather events impact aquaculture and fisheries through infrastructure damage
	Market	. Seafood as a major source of protein in Asian countries, with increasing global demand . Rising demand for Asian seafood-based cuisine driven by global food trends (e.g., sushi, raw fish, gyoza) . Strong global demand for shrimp . Development of responsible practices and sustainable management to meet the growing demand for high-quality products . Expansion of aquaculture by targeting high-value products and market demand (e.g., seaweed farming)	. Strong competition from neighboring countries (Vietnam, Indonesia, India) with lower labor costs and similar products . Dependence on competing countries for raw material supply (e.g., fishing gear, fishmeal) . High volatility in prices and demand . Vulnerability to global competition and demand fluctuations due to heavy reliance on exports
	Occupations	. Raising awareness among fishers and aquaculture producers about best practices	. Competition combined with rising raw material costs leading to reduced incomes . International regulations posing challenges for Thai fishers and aquaculture producers
	Policies	. Strengthening existing regulations (quotas, standards, etc.) . Implementing stricter measures against illegal fishing in Thai waters . Establishing traceability systems and quality certification schemes (e.g., MSC) —> Enhancing Thailand's competitiveness in the global market	. Lack of control leading to reduced demand from certain markets (e.g., Europe), which have increasingly strict quality requirements

10. Conclusion with a strategic perspective aimed at promoting more sustainable economic development of the sector

Fisheries and aquaculture play a central role in Thailand's economy and food security. However, numerous challenges must be addressed to transition toward more sustainable and environmentally responsible practices.

Thailand is currently making progress in addressing issues related to illegal, unreported, and unregulated (IUU) fishing, as well as overfishing and fleet overcapacity. By adopting a science-based fisheries management approach aligned with international standards, the country is implementing essential reforms. Nevertheless, further efforts are required to modernize the fishing fleet and techniques in order to promote more selective and sustainable practices, thereby allowing marine ecosystems to recover. Strengthening monitoring, control, and enforcement capacities therefore remains a key priority.

Two major strengths of Thailand are its food self-sufficiency, which enables the export of large quantities of food products, and its multi-species fishing fleet capable of harvesting significant volumes. Although tuna fisheries have not achieved the expected level of success, the country has successfully capitalized on the market by developing a strong canning industry, becoming a global leader in canned tuna. This strategy allows Thailand to import raw tuna, process it, and re-export it with substantial added value.

Aquaculture, for its part, provides an important source of food and protein but remains a fragile industry, highly vulnerable to disease outbreaks that can render products unsuitable for consumption. The development of sustainable practices such as Integrated Multi-Trophic Aquaculture (IMTA) could help mitigate these risks while improving the nutritional quality of production. This approach would also reduce environmental impacts by preserving ecosystems and limiting water pollution. Moreover, such solutions are economically viable and do not require advanced technologies, making them accessible for the country (Sampantamit et al., 2020).

Another challenge for the aquaculture sector lies in the implementation of stricter food safety and social standards, which may be difficult for small-scale family businesses to meet. However, with appropriate government support, these measures could foster a transition toward a quality-oriented sector, in line with the growing global demand for high-quality products.

Finally, improving working conditions in the aquaculture sector is essential. This includes the implementation of training programs, infrastructure development, and financial support to help workers cope with competition and economic and climate-related challenges. In addition, stronger enforcement measures are needed to combat forced labor and child labor, which undermine both social equity and the employment of local workers.

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